
GRAPHICS USER INTERFACE INTRO NOTES

Types of user interface :-

Command language-based interface.

- Menu-based UI.
- Natural language UI.
- Touch sensitive UI.
- Voice controlled UI.
- Web-based interface (UI).

Graphic User Interface (GUI)

The most widely used type of UI in use today is the Graphic User Interface (GUI). It accepts input through the keyboard, mouse and display the output on the user's computer screens. The action in a GUI usually performed through direct manipulation of the graphical elements.

The most common elements of a GUI include:

1. Window
2. Icon
3. Menu
4. Cursor pointer

Process of User Interface Design

1. Interface analysis.

2. Interface design.
3. Interface construction.
4. Interface implementation.
5. Interface validation.

****Interface Analysis****

- This is like studying how things are laid out and work on a website or an app.
- We look at what are these (like buttons, menus, text boxes)
- Where they are placed, and how they function.
- It's a something like taking apart a toy to see how it's made and how it works, but with website and apps instead.

****Interface Design****

- We decide how the interface will look and work based on what we learned during analysis.
- We choose colors, fonts, and shapes to make it look nice and easy to use.
- It's like a drawing a blueprint or sketch for a house before building it, but for a website or app.

****Interface Construction****

- This stage involves assembling all the elements needed for the interface based on the design.
- It's like putting together all the buttons, menus, and other components are in place and functioning correctly.
- Think of its as building a structure using the materials and plans provided by the design phase.

****Interface Implementation****

- Here, we actually build the interface based on the design we made.

- Developers write the code to make everything work as it should, like making buttons clickable and menus open when you tap them.
- It's like constructing a building according to the blueprint we draw.

****Interface Validation****

- After we've built interface, we test it to make sure it works well.
- We check if all the buttons do what they're supposed to, if the colors and fonts look good, and if it's easy for people to use.
- It's similar to trying out a new game to see if it's fun and works smoothly.

Principle and Attributes of User Interface Design

****Structure Principle****

This principle is concerned with overall UI architecture

The design of the interface should be visually, theoretically, and linguistically clear. (**Linguistic: Pertaining to language, include structure, meaning and usage**) User can easily navigate between different, parts of a website and quickly find important information or sections.

Tips/Techniques to support the structure principle

- You should group the logically connected items to communicate and separate unrelated items to achieve visual organization.
- Design page-specific navigation and access to information.
- Keep the screen less cluttered and easier to understand.
- Present the flow of actions, information, responses, and visual preparations in a sensible order that is easy to remember and place in context.

Parts of User Interface Design:-

The fundamental parts of most UIs are as follows:

- Input controls
- Navigational components
- Informational components
- Containers

Input controls:-

An input control defines the way in which the system captures information and it is the most interactive components of the UI. Some common interface input controls consist of elements such as buttons, radio buttons, check boxes, drop-down buttons, list boxes and text fields.

Here is a list of the input control elements-

1. Buttons
2. Radio buttons
3. Check box
4. Drop-down list
5. Drop-down button
6. List box
7. Text field

Navigational components:-

Navigational components provide a way for the users to tell the system what to do. Some of these includes search fields, breadcrumbs, pagination, tags, icons and image carousel.

List of the navigational component elements-

1. Search field
2. Breadcrumbs

3. Pagination
4. Tags
5. Icon
6. Image Carousel

Informational components:-

Some common interface information components are tooltips, notifications, progress bars, message boxes and modal window.

List of the informational component elements-

1. Tooltip
2. Notification
3. Progress Bar
4. Message Box
5. Modal-window (pop-up)

Containers:-

Containers group related content and maintain structure. The types of containers are cards, panels or sections, grids and lists.

List of the purpose of containers-

1. Organization
2. Clarity
3. Focus
4. Structure
5. Reusability

Introduction to UI & Visual Design Principles

UI (User Interface) and visual design principles help to create interfaces that are not only visually appealing but also easy to use. Understanding and applying these principles can lead to more intuitive and effective designs.

Principles of UI Design

Structure

Organize the interface in a meaningful and consistent way. Group related items together and separate unrelated items. Ensure the structure reflects the user's mental model and the task flow.

Simplicity

Keep the interface as simple as possible. Remove unnecessary elements and use clear, concise language. Simple interfaces are easier to learn and use.

Visibility

Make important elements visible and easily accessible. Ensure users can see the information they need when they need it, without cluttering the interface.

Feedback

Provide feedback to users about what is happening. This can include visual cues, sounds, or messages that indicate the result of their actions.

Tolerance

Design for error tolerance. Allow users to undo actions, and provide helpful error messages that guide users towards fixing issues.

Principles of Visual Design in UX

Balance

Balance refers to the distribution of visual weight within a design. A well-balanced design feels stable and aesthetically pleasing.

Contrast

Contrast is used to highlight important elements and create visual interest. It involves using different colors, shapes, and sizes to differentiate elements.

Scale

Scale involves using relative sizes to indicate importance and create a visual hierarchy. Larger elements are often perceived as more important.

Visual Hierarchy

Visual hierarchy is the arrangement of elements to guide the user's eye to the most important information first. It involves using size, color, and placement to prioritize content.

Gestalt Principles

Gestalt principles describe how people perceive visual elements as unified wholes. These principles include proximity, similarity, continuity, closure, and figure-ground.

Gestalt (German word Shape/Form) principles are:

Rules of human perception explaining how our brains naturally group visual elements into organized wholes, seeing patterns and unified forms rather than just separate parts, with key laws including **Proximity** (close items grouped), **similarity** (similar items grouped by shape/color), **Closure** (filling in gaps), **Continuity** (following smooth paths), and **Figure-Ground** (separating foreground from background). Designers use these principles to create intuitive, aesthetically pleasing, and easy-to-understand user interfaces and visuals.

{Perception: A particular way of looking at or understanding something; an opinion}

{Perceive: to become aware of, known, or identify by means of the senses to understand or think of something in a particular way}

Key Gestalt Principles:

- Proximity: Objects close to each other are perceived as a group.
- Similarity: Elements that share visual characteristics (color, shape, size) are seen as related.
- Closure: The mind fills in missing information to perceive complete shapes.
- Continuity: The eye follows smooth lines or curves, perceiving elements along them as connected.
- Common Fate: objects moving in the same direction are seen as a single unit.
- Connectedness: Visually linked elements (via lines, arrows) guide user attention.

How they're used:

- UI/UX Design: To structure content, group related features, and guide user attention.
- Data Visualization: To make complex data clearer by emphasizing patterns and relationships.

Model in User interface design

Four types of Model user interface design:

1. User model.
2. Design model.
3. Mental model.

4. Implementation model.

User Model:-

- This shows how the user thinks and behaves while using the system.
- It helps designers understand users needs, goals, and skills.
- Example: A beginner needs simple buttons and clear icons.

Design Model:-

- This is the designer's idea of how the system should work.
- It includes layout, color, structure, and functions.
- The goal is to make the design match what users expect.

Mental Model:-

- This is the user's understanding of how the system works.
- A good UI helps users predict what will happen when they click something.
- Example: A trash bin icon means delete.

Implementation Model:-

- This is how the system is actually built by programmers.
- It includes the technical structure and real working of the system.
- It should support the design so users can use it smoothly.

An implementation model is a representation of how a system actually works. It consist of the look and feel of the interface along with all supporting information such as resources, tutorials, help files, and videos that describes system syntax and semantics.